1. What is an Operating System? What are its primary functions?

Follow-up Questions:

- Why is an operating system necessary?
- Can a computer run without an OS?
- Name some popular operating systems.

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2. What is the difference between a Process and a Thread?

Follow-up Questions:

- Which one is lightweight?
- Do threads share memory?
- Which is faster to create?

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3. What are the different states of a Process?

Follow-up Questions:

- Explain each state.
- What happens during a context switch?
- Which state comes after the Ready state?

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4. What is CPU Scheduling?

Follow-up Questions:

- Why is scheduling required?
- Explain FCFS scheduling.
- Explain Round Robin scheduling.
- What is Time Quantum?

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5. What is Deadlock?

Follow-up Questions:

- What are the four necessary conditions for deadlock?

- How can deadlock be prevented?
- Give a real-life example.

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6. What is Paging?

Follow-up Questions:

- Why is paging used?
- What is a page?
- What is a frame?
- How is paging different from segmentation?

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7. What is Virtual Memory?

Follow-up Questions:

- Why do we need virtual memory?
- What is demand paging?
- What is a page fault?

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8. What is the difference between Multiprocessing and Multithreading?

Follow-up Questions:

- Which provides better isolation?
- Which consumes less memory?
- Give one practical example of each.

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9. What is Synchronization?

Follow-up Questions:

- Why is synchronization required?
- What is a Race Condition?
- What are Mutex and Semaphore?

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10. Scenario-Based Question

A banking application has multiple users transferring money between accounts simultaneously.

Questions:

- What problems can occur if synchronization is not used?
- Which synchronization technique would you use?
- What is a race condition in this scenario?
- How would you prevent data inconsistency?